

Experiment:-3

Aim: Details and Working of a Rainwater Detector for Protecting Clothes Using Arduino

Theory:

Rainwater Detector for Clothes Protection: Overview A Rainwater Detector is an automated system designed to detect rain and move clothes inside a shelter to prevent them from getting wet. The system operates using an Arduino Uno, a rain sensor, and a servo motor to retract the rod holding the clothes when rain is detected. This ensures the clothes remain dry and protected without manual intervention.

Key Components

1. Arduino Uno:

- Acts as the central controller, processing data from the rain sensor.
- Based on the sensor readings, it controls the servo motor to move the rod inside the shelter.

2. Rain Sensor (YL-83 or FC-37):

- Detects the presence of rain by measuring conductivity across its surface.
- Sends signals to the Arduino when raindrops are detected.
- Sensitivity can be adjusted to determine the level of precipitation required to trigger the system.

3. Servo Motor (SG90 or MG995):

- Controls the movement of the rod holding the clothes.
- The Arduino sends commands to rotate the servo motor, either retracting the rod inside the shelter or extending it outside.

4. Rod Mechanism:

- A rod holding the clothes, which moves in and out of the shelter based on rain detection.
- Designed to automatically retract inside the shelter when rain is detected.

How the Rainwater Detector Works

1. **Sensor Input:**
 - The rain sensor continuously monitors for raindrops.
 - When rain is detected, the sensor sends an analog signal to the Arduino.
2. **Decision Making:**
 - If the sensor detects rain (analog signal above a set threshold), the Arduino activates the servo motor to retract the rod inside the shelter.
 - If no rain is detected, the Arduino extends the rod outside for drying clothes.
3. **Motor Control:**
 - The Arduino sends a PWM signal to the servo motor to rotate it to the required angle.
 - **0°:** The rod is extended outside.
 - **90° or 180°:** The rod is retracted inside the shelter.
4. **Continuous Loop:**
 - The system continuously checks for rain and adjusts the rod position accordingly.
 - This ensures real-time response and automatic protection.

Simple Operational Flow

1. **Start:**
 - System initializes; Arduino, sensor, and servo motor are powered on.
2. **Measure Rain Presence:**
 - The rain sensor detects moisture levels and sends data to the Arduino.
3. **Decision Making:**
 - **If rain is detected:**
 - The Arduino signals the servo motor to retract the rod inside the shelter.
 - **If no rain is detected:**
 - The Arduino signals the servo motor to extend the rod outside.
4. **Repeat:**
 - The system continuously monitors rain and operates accordingly.

Example: Arduino Code for Basic Rain Detection and Clothes Protection

```
#define RAIN_SENSOR A0
#include <Servo.h>
Servo rodServo;

void setup() {
  pinMode(RAIN_SENSOR, INPUT);
  rodServo.attach(9);
  Serial.begin(9600);
}

void loop() {
  int rainValue = analogRead(RAIN_SENSOR);
  Serial.print("Rain Sensor Value: ");
  Serial.println(rainValue);

  if (rainValue > 500) { // Threshold for rain detection
    retractRod();
  } else {
    extendRod();
  }
  delay(500);
}

void extendRod() {
  rodServo.write(0); // Extend rod outside
}

void retractRod() {
  rodServo.write(90); // Retract rod inside the shelter
}
```

Conclusion: The Rainwater Detector for Protecting Clothes Using Arduino is an effective automated system that ensures clothes remain dry during rainfall. By integrating an Arduino Uno, a rain sensor, and a servo motor, the system detects rain in real-time and takes immediate action to retract the rod inside the shelter. This hands-free solution prevents water damage and showcases the practical application of Arduino-based automation. The concept can be expanded for larger-scale implementations, such as automated laundry drying systems.